A First Course in Causal Inference Ch. 11

Kodai Minesaki

Faculty of Engineering, The University of Tokyo

November 11, 2026

Ch. 12 The Doubly Robust or the Augmented Inverse Propensity Score Weighting Estimator for the Average Causal Effect

Ch. 12 The Doubly Robust or the Augmented Inverse Propensity Score Weighting Estimator for the Average Causal Effect

Assumption 12.0

Ignorability $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ and overlap $0 < e(X) < 1$.

Under Assum. 12.0, we have two identification formula of the average causal effect $\tau = \mathbb{E}[Y(1) - Y(0)]$. First, the outcome regression formula is

$$\tau = \mathbb{E}[\mu_1(X)] - \mathbb{E}[\mu_0(X)], \tag{1}$$

where,

$$\mu_1(X) = \mathbb{E}[Y(1) \mid X] = \mathbb{E}[Y \mid Z = 1, X]$$
$$\mu_0(X) = \mathbb{E}[Y(0) \mid X] = \mathbb{E}[Y \mid Z = 0, X].$$

Second, the IPW formula is

$$\tau = \mathbb{E}\left[\frac{ZY}{e(X)}\right] - \mathbb{E}\left[\frac{(1-Z)Y}{1-e(X)}\right], \tag{2}$$

where $e(X) = \mathbb{E}[Z = 1 \mid X]$. Eq. (**??**) is consistent if the outcome model is correctly specified, and Eq. (**??**) is consistent if the propensity score model is correctly specified.

- What if one of them is not correctly specified?
- We are then motivated to propose an estimator that is consistent if either of them is correctly specified.
- It is called the *doubly robust* estimator (**??**).

12.1 The doubly robust estimator

- We posit a model for the conditional means of the outcome $\mu_1(X, \beta_1), \mu_0(X, \beta_0)$, with parameters β_1, β_0.
- If the outcome model is corectly specified, then $\mu_1(X) = \mu_1(X, \beta_1)$ and $\mu_0(X) = \mu_0(X, \beta_0)$.
- We posit a model for the propensity score $e(X, \alpha)$, with a parameter α.
- If the propensity score model is correctly specified, then $e(X) = e(X, \alpha)$.
- In practice, both models may be misspecified.
- Sometimes, we call them *working models* due to the possibility of misspecification.

12.1 The doubly robust estimator

Define

$$\tilde{\mu}_1^{\mathsf{dr}} = \mathbb{E}\left[\frac{Z\{Y-\mu_1(X,\beta_1)\}}{e(X,\alpha)} + \mu_1(X,\beta_1)\right], \tag{3}$$

$$\tilde{\mu}_0^{\mathsf{dr}} = \mathbb{E}\left[\frac{(1-Z)\{Y-\mu_0(X,\beta_0)\}}{1-e(X,\alpha)} + \mu_0(X,\beta_0)\right], \tag{4}$$

which can be written as

$$\tilde{\mu}_1^{\mathsf{dr}} = \mathbb{E}\left[\frac{ZY}{e(X,\alpha)} - \frac{Z-e(X,\alpha)}{e(X,\alpha)}\mu_1(X,\beta_1)\right], \tag{5}$$

$$\tilde{\mu}_0^{\mathsf{dr}} = \mathbb{E}\left[\frac{(1-Z)Y}{1-e(X,\alpha)} - \frac{e(X,\alpha)-Z}{1-e(X,\alpha)}\mu(X,\beta_0)\right]. \tag{6}$$

12.1 The doubly robust estimator

- Eq. (**??**), (**??**) augment the outcome regression estimator by inverse propensity score weighting terms of the residuals $(Y - \mu_Z(X, \beta_Z))$.
- Eq. (**??**), (**??**) augment the IPW estimator by the imputed outcomes.
- For this reason, the doubly robust estimator is also called the *augmented inverse propensity score weighting* (AIPW) estimator.
- The augmentation strengthens the theoretical properties in the following sense.

Theorem 12.1

Assume ignorability $Z \perp\!\!\!\perp \{Y(1), Y(0)\} \mid X$ and overlap $0 < e(X) < 1$.

1. If either $e(X, \alpha) = e(X)$ or $\mu_1(X, \beta_1) = \mu_1(X)$, then $\tilde{\mu}_1^{\mathsf{dr}} = \mathbb{E}[Y(1)]$.
2. If either $e(X, \alpha) = e(X)$ or $\mu_0(X, \beta_1) = \mu_0(X)$, then $\tilde{\mu}_0^{\mathsf{dr}} = \mathbb{E}[Y(0)]$.
3. If either $e(X, \alpha) = e(X)$ or $\{\mu_1(X, \beta_1) = \mu_1(X), \mu_0(X, \beta_1) = \mu_0(X)\}$, then $\tilde{\mu}_1^{\mathsf{dr}} - \tilde{\mu}_0^{\mathsf{dr}} = \tau$.

12.1 The doubly robust estimator

- By Thm. 12.1, **if either the propensity score model or the outcome model is correctly specified,** $\tilde{\mu}_1^{\mathsf{dr}} - \tilde{\mu}_0^{\mathsf{dr}} = \tau$.

Proof.

$$
\begin{aligned}
\tilde{\mu}_1^{\mathsf{dr}} - \mathbb{E}[Y(1)] &= \mathbb{E}\left[\frac{Z\{Y(1) - \mu_1(X, \beta_1)\}}{e(X, \alpha)} + \mu_1(X, \beta_1) - Y(1)\right] \\
&= \mathbb{E}\left[\frac{Z - e(X, \alpha)}{e(X, \alpha)}(Y(1) - \mu_1(X, \beta_1))\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{Z - e(X, \alpha)}{e(X, \alpha)}(Y(1) - \mu_1(X, \beta_1)) \mid X\right]\right] \quad (\because \text{LIE}) \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{Z - e(X, \alpha)}{e(X, \alpha)} \mid X\right] \times \mathbb{E}\left[Y(1) - \mu_1(X, \beta_1) \mid X\right]\right] \quad (\because \text{strong ignorability}) \\
&= \mathbb{E}\left[\frac{e(X) - e(X, \alpha)}{e(X, \alpha)} \times \{\mu_1(X) - \mu_1(X, \alpha)\}\right].
\end{aligned}
$$

12.1 The doubly robust estimator

We can obtain their sample analogs to construct a doubly robust estimator for τ.

12.1 Definition (doubly robust estimator for the average causal effect)

Based on the data $(X_1, Z_i, Y_i)_{i=1}^n$, we can obtain a doubly robust estimator for τ by the following steps:

1. obtain the fitted values of the propensity scores: $e(X_i, \hat{\alpha})$;
2. obtain the fitted values of the outcome means: $\mu_1(X_i, \hat{\beta}_1), \mu_0(X_i, \hat{\beta}_0)$;
3. construct the doubly robust estimator: $\hat{\tau}^{dr} = \hat{\mu}_1^{\text{dr}} - \hat{\mu}_0^{\text{dr}}$, where

$$\hat{\mu}_1^{\text{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{Z_i\{Y_i - \mu_1(X_i, \hat{\beta}_1)\}}{e(X_i, \hat{\alpha})} + \mu_1(X_i, \hat{\beta}_1)\right],$$

$$\hat{\mu}_0^{\text{dr}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{(1 - Z_i)\{Y_i - \mu_0(X_i, \hat{\beta}_0)\}}{1 - e(X_i, \hat{\alpha})} + \mu_0(X_i, \hat{\beta}_0)\right]$$

12.1 The doubly robust estimator

By definition, we can also write the doubly robust estimator as:

$$\hat{\tau}^{\text{dr}} = \hat{\tau}^{\text{reg}} + \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i\{Y_i - \mu_1(X_i, \hat{\beta}_1)\}}{e(X, \hat{\alpha})} - \frac{1}{n}\sum_{i=1}^{n} \frac{(1 - Z_i)\{Y_i - \mu_0(X_i, \hat{\beta}_0)\}}{1 - e(X_i, \hat{\alpha})}.$$

We can also rewrite it as

$$\hat{\tau}^{\text{dr}} = \hat{\tau}^{\text{ipw}} - \frac{1}{n}\sum_{i=1}^{n} \frac{Z_i - e(X_i, \hat{\alpha})}{e(X_i, \hat{\alpha})}\mu_1(X_i, \hat{\beta}_1) + \frac{1}{n}\sum_{i=1}^{n} \frac{e(X_i, \hat{\alpha}) - Z_i}{1 - e(X_i, \hat{\alpha})}\mu_0(X_i, \hat{\beta}_0).$$

12.2 More intuition and theory for the doubly robust estimator

The IPW estimator for μ_1 based on $\mu_1 = \mathbb{E}\left[ZY/e(X)\right]$ completely ignores the outcome model of Y. With possibly misspecified outcome model $\mu_1(X, \beta_1)$,

$$\mu_1 = \mathbb{E}[Y(1)] = \mathbb{E}[Y(1) - \mu_1(X, \beta_1)] + \mathbb{E}[\mu_1(X, \beta_1)].$$

If we apply the IPW formula to the first term in the above formula:

$$\mu_1 = \mathbb{E}\left[\frac{Z(Y(1) - \mu_1(X, \beta_1))}{e(X)}\right] + \mathbb{E}[\mu_1(X, \beta_1)] = \mathbb{E}\left[\frac{Z(Y(1) - \mu_1(X, \beta_1))}{e(X)} + \mu_1(X, \beta_1)\right],$$

which holds if the propensity core model correct without assuming that the outcome model is correct.

Future Reading

- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. https://academic.oup.com/ectj/article/21/1/C1/5056401

References I

Bang, H. and J. M. Robins (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics 61*(4), 962–973.

Scharfstein, D. O., A. Rotnitzky, and J. M. Robins (1999). Adjusting for nonignorable drop-out using semiparametric nonresponse models. *Journal of the American Statistical Association 94*(448), 1096–1120.